

ML Classification Models - Assignment 2

a. Problem statement

The objective of this assignment is to implement and compare multiple classification models to predict whether an individual earns more than \$50K per year using a public classification dataset.

The app is built in Streamlit and allows users to upload a CSV test file, select a model, and view evaluation metrics, confusion matrix, and classification report.

b. Dataset description

The chosen dataset is the Adult (Census Income) dataset, a binary classification dataset downloaded from OpenML/UCI.

Dataset details:

- Type: Binary classification
- Number of instances: More than 500 (Adult dataset has 48,842 rows)
- Number of features: More than 12 features
- Target variable: `income`
- Target labels: `<=50K` and `>50K`

The dataset contains demographic and employment-related features such as age, education, marital status, occupation, hours per week, capital gain/loss, and native country.

c. GitHub Repository Link

GitHub Repository Link: <https://github.com/Akassh09/ml-streamlit-app.git>

Live Streamlit App Link

Streamlit App: <https://ml-app-app-gqzpw3uv4eowkprxcgghr.streamlit.app/>

d. Models used

The following six classification models were implemented and evaluated on the same dataset:

- Logistic Regression
- Decision Tree Classifier
- K-Nearest Neighbor Classifier
- Naive Bayes Classifier
- Random Forest Classifier (Ensemble)

Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.84599	0.90465	0.73159	0.59813	0.65816	0.56479
Decision Tree	0.80597	0.74733	0.60393	0.63113	0.61723	0.48757
kNN	0.82488	0.85203	0.66566	0.58965	0.62535	0.51323

Naive Bayes	0.64478	0.85068	0.40464	0.91882	0.56184	0.41197
Random Forest (Ensemble)	0.84456	0.89978	0.72093	0.60839	0.65989	0.56347

Observations on model performance

ML Model Name	Observation about model performance
Logistic Regression	Performed best overall with strong balance between accuracy, AUC, precision, and MCC. Good generalization for the chosen dataset.
Decision Tree	Simple and interpretable, but overfitting risk is visible and performance is weaker than logistic regression and random forest.
kNN	Reasonably good accuracy, but it is sensitive to feature scaling and showed slightly lower precision/recall balance compared to logistic regression.
Naive Bayes	Very high recall but poor precision and lower accuracy, which indicates it tends to classify more samples as positive.
Random Forest (Ensemble)	Strong performance, close to logistic regression, and robust due to ensemble averaging; good precision and F1 score.

Overall Winner for the dataset

Overall Winner: Logistic Regression

Reason: It achieved the highest AUC and the best overall trade-off among Accuracy, Precision, F1, and MCC on this dataset.

How to run the project

1. Install the dependencies:

```
pip install -r requirements.txt
```

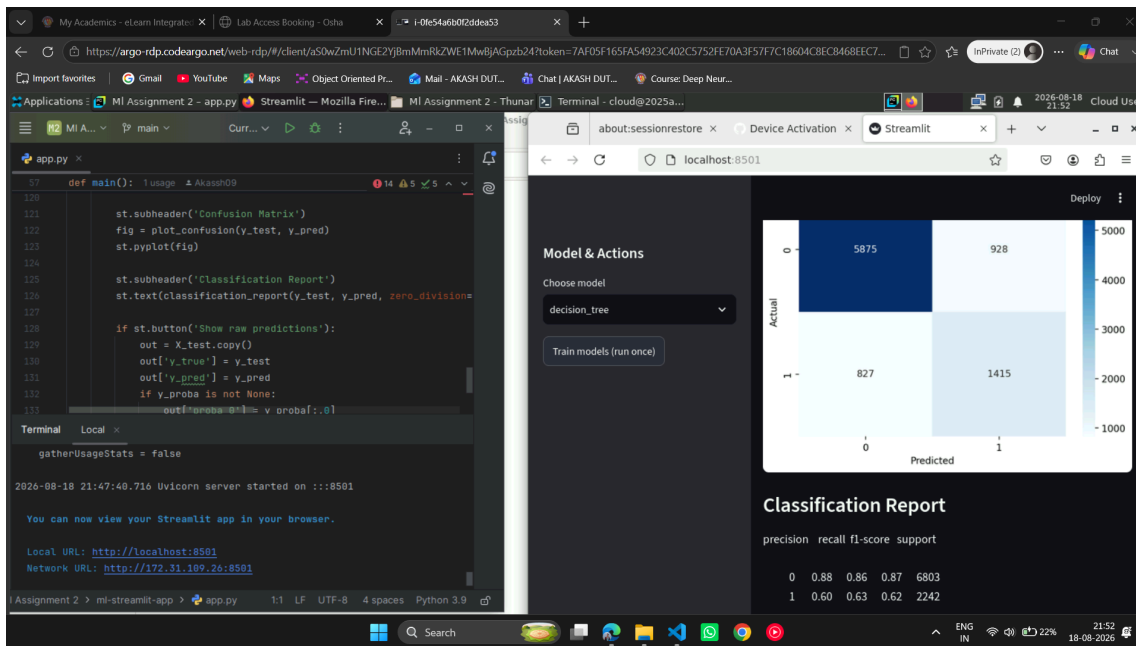
2. Train the machine learning models:

```
python train_models.py
```

3. Run the Streamlit application:

```
streamlit run app.py
```

Screenshot



Project files

- `app.py` — Streamlit dashboard
- `train_models.py` — model training and dataset handling
- `requirements.txt` — required Python packages
- `model/` — saved trained models
- `test_data.csv` — sample test dataset
- `README.md` — project documentation